STA 135 Midterm 2 Project

Maxime Pouokam
Names: Anishka Dhar, Weitao Chen, Ian Dang



## I. Introduction

This report is investigating which characteristics of houses are most associated with their sale price in King County. Having an understanding of what factors affect housing prices is important for buyers, sellers, realtors, and even policymakers since it provides crucial information regarding the housing market. Knowing the influential factors could give further insight into the market trends as well as consumer preference. The question we are looking to research is: which housing characteristics can significantly predict the sale price of a house ? (as well as how well they can explain the variation in prices). In order to answer this, we will be taking a multiple linear regression modeling approach to look at the relationships between price and other variables.

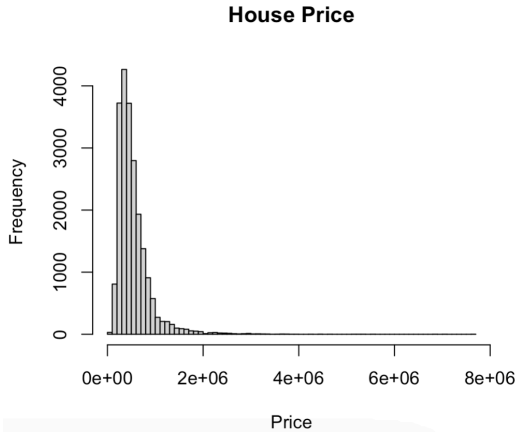
## II. Summary

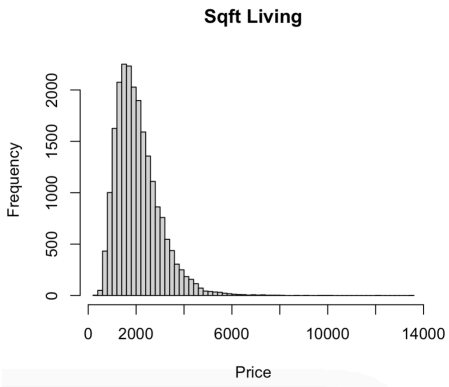
The dataset includes 21,613 property sales from King County Washington with the response variable price (sale price of each house). There are 20 predictor variables that describe size, quality, age, condition, and location.

Before modeling, some data cleaning was required. Id and date columns were removed, since they are unnecessary for modeling. Missing values in the year renovated were replaced with 0, indicating no recorded renovation. Missing values in view and waterfront variable were additionally both replaced with 0.

Summary statistics suggest high variation in housing factors. Houses have from 1 to 33 bedrooms, with living areas from 370 to 13540 square feet. The construction grades range from 3 to 13, with a median of 7 suggesting that houses tend to be average quality.

The response variable price varies significantly throughout the set, and ranges from as low as $78,000 to $7,700,000. The mean price is $540,297, and the median is $450,000. Looking at the distribution of house prices, we can see that it is highly right skewed. This indicates that there is a much smaller number of very expensive homes, and



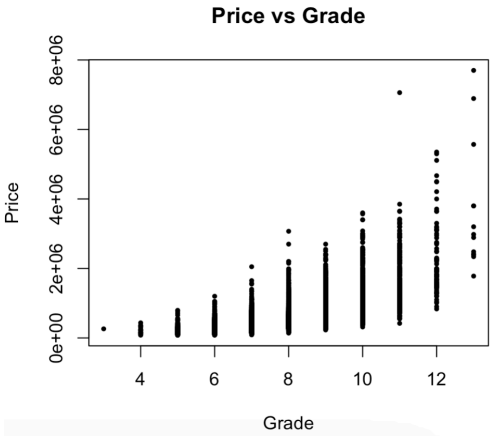






that there is significant variability across the data.

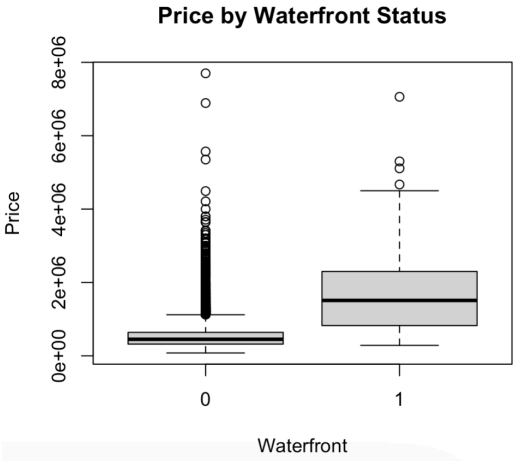
The next figure is the distribution of the living area (sqft_living variable). Most homes have living areas between 1000 and 3000 square feet, with a very small number of large homes as seen in the right skewed histogram.

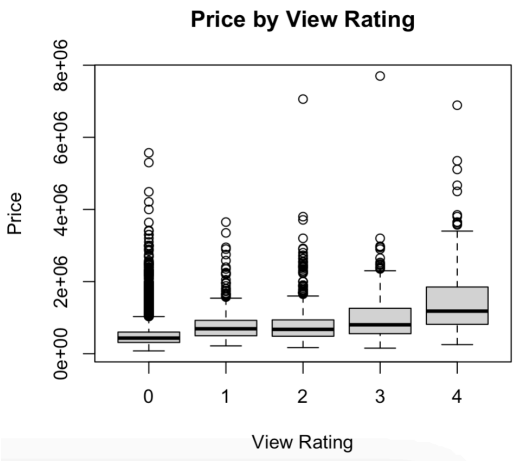
Next we have a scatter plot between the sale price and living area (sqft_living). We can see that there is a strong positive correlation between the two variables, suggesting that larger houses will tend to sell for higher prices. Additionally we find that the correlation coefficient between the two variables is 0.706 (which is one of the strongest correlations with price), further supporting this claim.

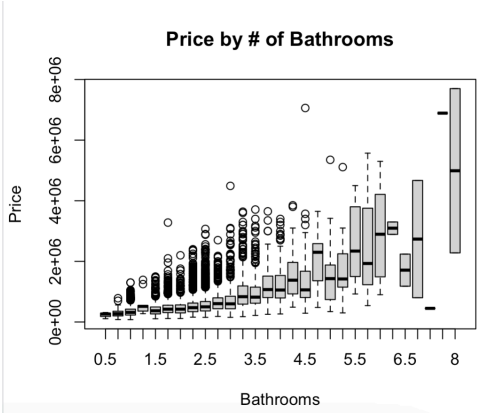



Next we can look at the relationship between price and construction grade. Here we can see that higher grades tend to belong to homes with higher prices, suggesting that construction quality is an important factor of house values. Additionally the correlation between price and construction grade is 0.664, which is again one of the stronger correlations, further supporting this claim.

Below are boxplots comparing price to waterfront and view rating. Waterfront houses seem to have higher prices than non waterfront houses, however it is not seemingly strong. Similarly homes with better views tend to be more expensive but with multiple outliers. The positive, but low, correlations of waterfront and view ratings for price (0.274 and 0.397 respectively) further support these findings.






Here we have the prices by the number of bathrooms. Homes with more bathrooms tend to have higher prices, with the median prices increasing as the number of bathrooms increase. There is also an increase in variability of price with an increase of bathroom numbers. There are some high priced outliers which become significant among houses with more than 4 bathrooms. Additionally the correlation of 0.526 further supports the claim of a strong association between bathroom and price.

Looking at the correlation matrix helps identify predictors that are strongly associated with sale price. The strongest of those are shown below.

| Variable | Correlation |
|---|---|
| sqft_living | 0.706 |
| grade | 0.664 |
| sqft_above | 0.612 |
| sqft_living15 | 0.582 |
| bathrooms | 0.526 |

An analysis of correlation also suggests a chance of multicollinearity. For example sqft_living is strongly correlated with sqft_above, grade, and sqft_living15 (0.876, 0.764, and 0.757 respectively). Grade is also strongly correlated with sqft_above and sqft_living15 (0.758 and 0.717). These values suggest that there may be some overlap between predictors.

## III. Model Selection

The main goal of this analysis is to identify the housing characteristics that best explains the variation in King County house sales prices. Since the dataset contains many potential predictions related to home size, quality, age, and geographic location, multiple regression modeling was appropriate because it allows the simultaneous evaluation of several predictors while controlling the effects of the others.

The response variable in this analysis was the final sale price of the home. Candidate predictor variables included structural characteristics such as bedrooms, bathrooms, square footage of living space, lot size, and construction grade. Further predictor variables include location and neighborhood factors such as view quality, longitude, latitude, and distance from

downtown Seattle. Variables also include nearby homes, such as neighborhood living area and neighborhood lot size, which were included because surrounding property characteristics may influence local market value. The initial regression model includes all predictors that are believed to be important for explaining housing prices. A stepwise selection method using AIC then applied to simplify the model. This process removed variables that did not significantly improve the model while keeping the predictors that were most useful. Lower AIC values were preferred because they indicated a better balance between model accuracy and avoiding overfitting.

```
Call:
lm(formula = price ~ bedrooms + bathrooms + sqft_living + sqft_lot +
    waterfront + view + condition + grade + age + renovated +
    dist_downtown, data = kc)

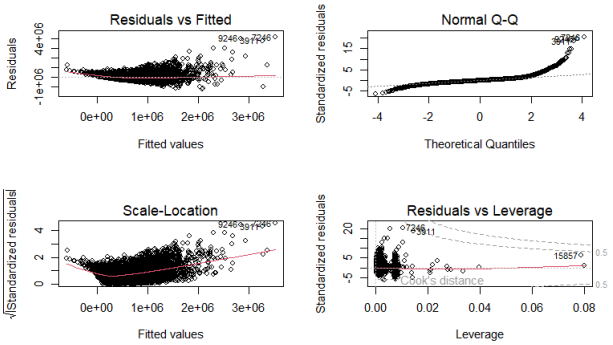
Residuals:
     Min       1Q   Median       3Q      Max
-1281784  -106553   -11895    85215  4176060

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   -7.038e+05  1.734e+04 -40.592  < 2e-16 ***
bedrooms      -3.729e+04  1.940e+03 -19.219  < 2e-16 ***
bathrooms      3.195e+04  3.221e+03   9.920  < 2e-16 ***
sqft_living    1.907e+02  3.161e+00  60.342  < 2e-16 ***
sqft_lot       1.823e-01  3.596e-02   5.071 4.00e-07 ***
waterfront     6.203e+05  1.856e+04  33.414  < 2e-16 ***
view           4.074e+04  2.104e+03  19.362  < 2e-16 ***
condition      2.622e+04  2.362e+03  11.103  < 2e-16 ***
grade          1.102e+05  2.047e+03  53.831  < 2e-16 ***
age            1.929e+03  7.285e+01  26.485  < 2e-16 ***
renovated      4.656e+04  8.117e+03   5.736 9.84e-09 ***
dist_downtown -6.518e+05  1.371e+04 -47.548  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 206300 on 21585 degrees of freedom
Multiple R-squared:  0.6849,    Adjusted R-squared:  0.6848
F-statistic:  4266 on 11 and 21585 DF,  p-value: < 2.2e-16
```

In addition to the main regression model, several focused regression models were fit to answer specific research questions related to bathrooms, construction grade, living space, lot size, and distance from downtown Seattle. Interaction terms were also considered when there was reason to believe that the effect of one predictor depended on another predictor. In particular, an interaction between waterfront status and view quality was included to determine whether scenic views have different pricing effects for waterfront and non waterfront homes. This follows the multiple regression framework discussed in class, where interaction terms are included as products of predictors to model joint effects.

## IV. Diagnostics



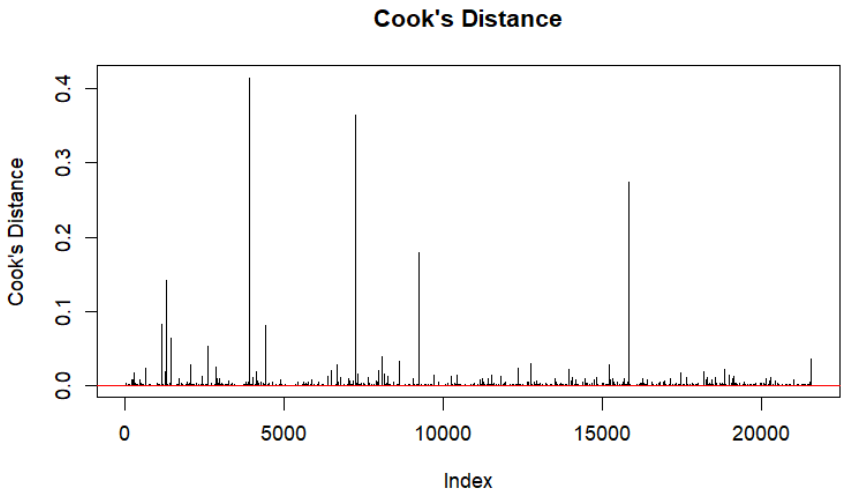

After selecting the regression model, diagnostic procedures were conducted to evaluate whether the assumptions of multiple linear regression were satisfied. The assumptions include linearity, constant variance of residuals, independence of observations, approximate normality of residuals, and outliers.

Residual versus fitted value plots were examined to evaluate whether the relationship between the predictors and the response variable was approximately linear and whether the variance of the residuals remained relatively constant across different fitted values. The residual

plots indicated some increase in variability for extremely high priced homes. This is expected in housing datasets since luxury homes tend to have much larger price variation than average homes. However, the overall linear trend remained reasonable for modeling purposes.

Normal Q-Q plots were also examined to evaluate the distribution to evaluate the distribution of residuals. While minor deviation from normality appeared in the tails of the distribution, especially for extremely expensive properties, the residuals were approximately normal overall. Since the dataset contains more than 21,000 observations, small deviations from normality are less concerning due to large sample size and asymptotic properties of least squares estimate.

Influential observations were evaluated using Cook's distance;. Several extremely expensive waterfront homes show large influence on the fitted regression model. Observations with Cook's distance greater than 4/n are removed before refitting the final model to reduce extreme skewed points.




Multicollinearity among predictors was assessed using variance inflation factors (VIF). some degree of multicollinearity was expected because housing variables such as bedrooms, bathrooms, square footage, and construction grade are naturally related. However, the VIF values did not indicate multicollinearity, showing that the regression coefficients are reasonably stable and interpretable.

The Breusch Pagan test was used to evaluate heteroskedasticity. The test suggested evidence of nonconstant variance in the residuals, which is common in real estate data because higher valued homes tend to exhibit greater variability in sales prices. Although this violates the constant variance assumption to some extent, the regression model still provides interpretation and predictive value since the sample size is large.

## V. Analysis

The final multiple linear regression model explained a substantial proportion of the variability in King County housing prices. The fitted regression model achieved an R^2 value and an adjusted R^2 value of approximately 0.685. This indicates that nearly 68.5% of the variation in housing prices can be explained by the selected predictor variables. The adjusted and ordinary R^2 are close, suggesting that the included predictors contributed meaningful explanatory information without excessive overfitting.

The overall significance of the regression model was evaluated using the global ANOVA F-test. The hypothesis for the test were:

$$H_0: \beta_1 = \beta_2 \ldots = \beta_p = 0$$

$$H_A: \text{At least one } \beta_j \neq 0$$

The fitted model produced an F-statistic of approximately 3910.66 with an extremely small p-value ($p < 0.001$). Therefore the null hypothesis was rejected, provided strong evidence that at least one predictor variable contributes significantly to explaining variation in housing prices.

*Individual Regression Coefficients*

For each predictor variable, individual t-tests were conducted to evaluate the hypotheses

$$H_0: \beta_j = 0$$

$$H_A: \beta_j \neq 0$$

These tests determine whether each predictor contributes significantly to the regression model after accounting for the remaining predictors.

```
Call:
lm(formula = price ~ bathrooms + bedrooms + sqft_living + grade +
    condition + age + waterfront + view, data = kc_final)

Residuals:
    Min      1Q  Median      3Q     Max
-480411  -95175   -5461   83248  744914

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.163e+05  1.207e+04 -67.623   <2e-16 ***
bathrooms    4.713e+04  2.347e+03  20.081   <2e-16 ***
bedrooms    -2.817e+04  1.486e+03 -18.961   <2e-16 ***
sqft_living  1.185e+02  2.407e+00  49.234   <2e-16 ***
grade        1.172e+05  1.495e+03  78.387   <2e-16 ***
condition    1.404e+04  1.679e+03   8.361   <2e-16 ***
age          2.951e+03  4.618e+01  63.909   <2e-16 ***
waterfront   6.649e+05  3.746e+04  17.748   <2e-16 ***
view         4.225e+04  1.688e+03  25.032   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 143100 on 20421 degrees of freedom
Multiple R-squared:  0.6746,    Adjusted R-squared:  0.6745
F-statistic:  5292 on 8 and 20421 DF,  p-value: < 2.2e-16

                   2.5 %       97.5 %
(Intercept) -839974.9835 -792652.4121
bathrooms     42527.1540   51727.3378
bedrooms     -31086.5742  -25261.7410
sqft_living     113.7679     123.2021
grade        114251.8239  120112.1461
condition     10747.5861   17330.1929
age            2860.8964    3041.9352
waterfront   591435.1883  738291.6567
view          38944.5077   45561.6189
```

*Bathrooms*

The estimated regression coefficient for bathrooms was approximately 47,130. This indicates that, after controlling for the remaining housing characteristics, one additional bathroom is associated with an expected increase of approximately $47,130 in housing price. The coefficient was statistically significant with a p value of less than 0.001. Showing strong evidence that bathrooms contribute to explaining variation in housing prices.
This result is economically reasonable because additional bathrooms improve household convenience and functionality, especially in larger homes. Bathrooms may also serve as a proxy for higher end housing construction and more modern home layouts.

*Living Space*

The coefficient associated with sqft_living was approximately 125.1, meaning that each additional square foot of interior living space increased the expected sale price by approximately $125 on average, holding the remaining predictors fixed. The corresponding p value was less than 0.001, indicating that living space is one of the strongest predictors of housing value. Additionally. The standard coefficient for sqft_living is approximately 0.404. Suggesting that living space is among the strongest predictors in the regression model after accounting for differences in measurement scales across variables.

```
Call:
lm(formula = price ~ grade + sqft_living + bathrooms + bedrooms +
    condition + age, data = kc_final)

Residuals:
    Min      1Q  Median      3Q     Max
-498873  -96536   -6172   83701 1022993

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.559e+05  1.231e+04 -69.523   <2e-16 ***
grade        1.209e+05  1.529e+03  79.043   <2e-16 ***
sqft_living  1.251e+02  2.458e+00  50.892   <2e-16 ***
bathrooms    4.996e+04  2.408e+03  20.750   <2e-16 ***
bedrooms    -3.160e+04  1.521e+03 -20.773   <2e-16 ***
condition    1.492e+04  1.724e+03   8.654   <2e-16 ***
age          3.148e+03  4.688e+01  67.148   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 147000 on 20423 degrees of freedom
Multiple R-squared:  0.657,     Adjusted R-squared:  0.6569
F-statistic:  6520 on 6 and 20423 DF,  p-value: < 2.2e-16


Call:
lm(formula = price ~ grade + sqft_living + bathrooms + bedrooms +
    condition + age, data = kc_final)

Standardized Coefficients::
(Intercept)       grade sqft_living   bathrooms    bedrooms   condition         age
         NA  0.52483990  0.40380185  0.14337731 -0.11059186  0.03837847  0.36545739
```

This result demonstrates that usable interior living space is highly valued in the Seattle housing market. Larger homes generally provide more functionality and flexibility, which substantially increases market demand and expected sale price.


```
Call:
lm(formula = price ~ grade + sqft_living + bathrooms + bedrooms +
    condition + age, data = kc_final)

Residuals:
    Min      1Q  Median      3Q     Max
-498873  -96536   -6172   83701 1022993

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.559e+05  1.231e+04 -69.523   <2e-16 ***
grade        1.209e+05  1.529e+03  79.043   <2e-16 ***
sqft_living  1.251e+02  2.458e+00  50.892   <2e-16 ***
bathrooms    4.996e+04  2.408e+03  20.750   <2e-16 ***
bedrooms    -3.160e+04  1.521e+03 -20.773   <2e-16 ***
condition    1.492e+04  1.724e+03   8.654   <2e-16 ***
age          3.148e+03  4.688e+01  67.148   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 147000 on 20423 degrees of freedom
Multiple R-squared:  0.657,     Adjusted R-squared:  0.6569
F-statistic:  6520 on 6 and 20423 DF,  p-value: < 2.2e-16


Call:
lm(formula = price ~ grade + sqft_living + bathrooms + bedrooms +
    condition + age, data = kc_final)

Standardized Coefficients::
(Intercept)       grade sqft_living   bathrooms    bedrooms   condition         age
         NA  0.52483990  0.40380185  0.14337731 -0.11059186  0.03837847  0.36545739
```


*Construction Grade*

The coefficient associated with construction grade was approximately 120,900, suggesting that a one unit increase in construction grade increased expected housing price by approximately $120,900 after controlling for other variables. The variable was highly statistically significant with a p value less than 0.001.

The 95% confidence interval indicates that the true average increase in housing price associated with a one unit increase in grade is likely between approximately $117,900 and $123,900. The standardized coefficient for grade was around 0.525, which was larger than the standardized coefficient for sqft_living. This suggested

that construction quality contributes more strongly to housing prices than additional living space after accounting for differences in variables.

This result indicates that buyers place substantial value on housing quality, materials, workmanship, and overall design characteristics. Construction grade captures many desirable housing traits that are not fully reflected by square footage alone.

*Lot Size*

The estimated coefficient for sqft_lot was approximately -0.184. Although statistically significant with a t statistic of approximately −5.10 and a p value less than 0.001, the magnitude of the effect was substantially smaller than the effect associated with interior living space.

The standardized coefficient for sqft_lot was approximately -0.021, indicating that lot size contributed very little to explaining variation in housing prices after controlling for the remaining housing characteristics. This suggests that buyers in the Seattle housing market place substantially greater value on usable interior living space than on total land area.

```
Call:
lm(formula = price ~ sqft_living + sqft_lot + grade + bathrooms +
    bedrooms + condition + age, data = kc_final)

Residuals:
    Min      1Q  Median      3Q     Max
-495044  -96481   -5936   83791 1019389

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.526e+05  1.232e+04 -69.205  < 2e-16 ***
sqft_living  1.274e+02  2.498e+00  51.002  < 2e-16 ***
sqft_lot    -1.839e-01  3.605e-02  -5.099 3.44e-07 ***
grade        1.207e+05  1.529e+03  78.939  < 2e-16 ***
bathrooms    4.902e+04  2.413e+03  20.313  < 2e-16 ***
bedrooms    -3.222e+04  1.525e+03 -21.129  < 2e-16 ***
condition    1.505e+04  1.723e+03   8.734  < 2e-16 ***
age          3.139e+03  4.689e+01  66.948  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 146900 on 20422 degrees of freedom
Multiple R-squared:  0.6574,    Adjusted R-squared:  0.6573
F-statistic:  5599 on 7 and 20422 DF,  p-value: < 2.2e-16


Call:
lm(formula = price ~ sqft_living + sqft_lot + grade + bathrooms +
    bedrooms + condition + age, data = kc_final)

Standardized Coefficients::
(Intercept) sqft_living    sqft_lot       grade   bathrooms    bedrooms   condition         age
         NA  0.41127108 -0.02140613  0.52399255  0.14067970 -0.11278139  0.03871655  0.36440793
```

*Distance from Downtown Seattle*

Distance from downtown Seattle showed a strong negative relationship with housing prices. The estimated coefficient for dist_downtown was approximately −529,200, indicating that homes located farther from downtown Seattle tended to have lower expected housing prices after controlling for other structural housing characteristics. The coefficient was highly statistically significant with a p value less than 0.001, providing strong evidence that proximity to downtown areas influences housing value.

Economically, this finding is reasonable because homes located closer to downtown Seattle generally

```
Call:
lm(formula = price ~ dist_downtown + sqft_living + grade + bathrooms +
    bedrooms + condition + age, data = kc_final)

Residuals:
    Min      1Q  Median      3Q     Max
-466619  -89944   -6128   81392  963106

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   -6.255e+05  1.213e+04  -51.59   <2e-16 ***
dist_downtown -5.292e+05  9.275e+03  -57.05   <2e-16 ***
sqft_living    1.515e+02  2.329e+00   65.04   <2e-16 ***
grade          1.062e+05  1.443e+03   73.60   <2e-16 ***
bathrooms      3.192e+04  2.258e+03   14.14   <2e-16 ***
bedrooms      -3.102e+04  1.413e+03  -21.96   <2e-16 ***
condition      2.221e+04  1.606e+03   13.83   <2e-16 ***
age            1.885e+03  4.885e+01   38.59   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 136500 on 20422 degrees of freedom
Multiple R-squared:  0.7042,    Adjusted R-squared:  0.7041
F-statistic:  6944 on 7 and 20422 DF,  p-value: < 2.2e-16

                   2.5 %       97.5 %
(Intercept)   -649290.930 -601754.6954
dist_downtown -547347.867 -510988.0165
sqft_living       146.930     156.0615
grade          103386.647  109043.6547
bathrooms       27497.904   36351.1253
bedrooms       -33789.954  -28252.1818
condition       19059.840   25355.3024
age              1789.084    1980.5684
```

provide better access to employment centers, transportation networks, restaurants, entertainment, and other urban amenities. As distance from the city center increases, housing demand and market values tend to decrease on average.

*Waterfront and View Effects*

Waterfront status and view quality both displayed positive relationships with housing prices. Holding the remaining predictors constant, waterfront homes were associated with an estimated increase of approximately $869,300 in expected sale price relative to non waterfront homes. In addition, each one unit increase in view quality was associated with an estimated increase of approximately $38,950 in housing price. Both predictors were statistically significant at the 0.05 level.

The interaction term between waterfront status and view quality was estimated at approximately −49,990, but the coefficient was not statistically significant (p = 0.620). This suggests there is insufficient evidence to conclude that the effect of view quality on housing price differs meaningfully between waterfront and non waterfront homes after controlling for the remaining housing characteristics.

```
Call:
lm(formula = price ~ waterfront * view + sqft_living + grade +
    bathrooms + bedrooms + condition + age + dist_downtown, data = kc_final)

Residuals:
    Min      1Q  Median      3Q     Max
-451356  -88682   -5546   80340  686533

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)     -5.921e+05  1.185e+04 -49.962   <2e-16 ***
waterfront       8.693e+05  3.914e+05   2.221   0.0264 *
view             3.895e+04  1.566e+03  24.868   <2e-16 ***
sqft_living      1.450e+02  2.278e+00  63.625   <2e-16 ***
grade            1.030e+05  1.408e+03  73.157   <2e-16 ***
bathrooms        2.952e+04  2.198e+03  13.433   <2e-16 ***
bedrooms        -2.782e+04  1.378e+03 -20.192   <2e-16 ***
condition        2.127e+04  1.562e+03  13.615   <2e-16 ***
age              1.721e+03  4.784e+01  35.975   <2e-16 ***
dist_downtown   -5.210e+05  9.026e+03 -57.716   <2e-16 ***
waterfront:view -4.999e+04  1.009e+05  -0.496   0.6201
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 132700 on 20419 degrees of freedom
Multiple R-squared:  0.7203,    Adjusted R-squared:  0.7201
F-statistic:  5257 on 10 and 20419 DF,  p-value: < 2.2e-16

                      2.5 %       97.5 %
(Intercept)     -615300.6312 -568844.9079
waterfront       102053.1217 1636534.0608
view              35883.9021   42024.7148
sqft_living         140.4957     149.4273
grade            100245.2216  105764.8241
bathrooms         25212.5530   33827.2140
bedrooms         -30522.0104  -25120.7542
condition         18205.9978   24329.6921
age                1627.1879    1814.7211
dist_downtown   -538662.3643 -503277.2148
waterfront:view -247670.6270  147688.8924
```

The results indicate that both waterfront location and scenic views independently contribute to higher housing prices. Waterfront properties likely command substantial premiums because of their scarcity, aesthetic appeal, and access to desirable amenities, while higher quality views increase perceived housing desirability and market value.

*Structural traits*

Nested regression models were compared using R 2 , AIC, and partial F tests. The full regression model achieved an R 2 value of approximately 0.657, indicating that approximately 65.7% of the variation in housing prices was explained by the included predictors. Structural housing variables such as living space, construction grade, bathrooms, condition, and waterfront status explained substantially more variation in housing prices than age related variables alone.

Although home age was statistically significant with a p value less than 0.001, its explanatory contribution was relatively small compared to structural and location related variables. The standardized coefficient for age was approximately 0.365, while the standardized coefficients for construction grade and living space were approximately 0.525 and 0.404 respectively. The estimated coefficient for age was positive, suggesting that some older homes may still command high prices because they are located in desirable neighborhoods or have undergone substantial renovations. Overall, the regression analysis suggests that structural housing quality, usable interior living space, geographic location, and waterfront amenities were among the most important determinants of housing prices in King County.

```
Call:
lm(formula = price ~ grade + sqft_living + bathrooms + bedrooms +
    condition, data = kc_final)

Residuals:
    Min      1Q  Median      3Q     Max
-494723 -109342  -13071   91007 1039339

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -5.935e+05  1.290e+04 -46.011  < 2e-16 ***
grade        9.679e+04  1.642e+03  58.938  < 2e-16 ***
sqft_living  1.504e+02  2.684e+00  56.054  < 2e-16 ***
bathrooms   -1.656e+04  2.425e+03  -6.829 8.82e-12 ***
bedrooms    -2.621e+04  1.678e+03 -15.621  < 2e-16 ***
condition    5.210e+04  1.803e+03  28.887  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 162400 on 20424 degrees of freedom
Multiple R-squared:  0.5813,    Adjusted R-squared:  0.5812
F-statistic:  5671 on 5 and 20424 DF,  p-value: < 2.2e-16


Call:
lm(formula = price ~ age, data = kc_final)

Residuals:
    Min      1Q  Median      3Q     Max
-409236 -183950  -56899  120410 1905456

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 527189.71    3154.94  167.10   <2e-16 ***
age           -665.56      60.09  -11.08   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 250100 on 20428 degrees of freedom
Multiple R-squared:  0.00597,   Adjusted R-squared:  0.005921
F-statistic: 122.7 on 1 and 20428 DF,  p-value: < 2.2e-16


Call:
lm(formula = price ~ grade + sqft_living + bathrooms + bedrooms +
    condition + age, data = kc_final)

Residuals:
    Min      1Q  Median      3Q     Max
-498873  -96536   -6172   83701 1022993

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -8.559e+05  1.231e+04 -69.523   <2e-16 ***
grade        1.209e+05  1.529e+03  79.043   <2e-16 ***
sqft_living  1.251e+02  2.458e+00  50.892   <2e-16 ***
bathrooms    4.996e+04  2.408e+03  20.750   <2e-16 ***
bedrooms    -3.160e+04  1.521e+03 -20.773   <2e-16 ***
condition    1.492e+04  1.724e+03   8.654   <2e-16 ***
age          3.148e+03  4.688e+01  67.148   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 147000 on 20423 degrees of freedom
Multiple R-squared:  0.657,     Adjusted R-squared:  0.6569
F-statistic:  6520 on 6 and 20423 DF,  p-value: < 2.2e-16
```

## VI. Interpretation

The result of the multiple regression analysis suggests that several housing characteristics significantly influence home sale prices in King County. Particularly, interior living space, construction grade, waterfront status, and proximity to downtown Seattle were among the strongest predictors of housing prices after controlling for the remaining variables in the model.

The regression coefficients show that homes with larger living areas tend to sell for significantly higher prices. Taking the effects of other predictors into account, each additional square foot of living space was tied to an increase of approximately $125 in expected sale price, suggesting that buyers place a considerable value on housing quality, materials, and overall design.

Bathrooms also showed a significant positive relationship with housing prices. Holding the remaining variables constant, an extra bathroom increased the expected sale price by approximately $47,000. This suggests that functionality and convenience are important factors to housing demand in the Seattle market.

Location variables were also highly influential. Distance from downtown Seattle showed a strong negative relationship with price, showing that homes farther from the city center generally sell for lower prices. This result is economically reasonable because homes closer to downtown areas usually provide better access to employment centers, transportation, entertainment, and other convenient amenities. Waterfront properties were associated with substantially higher prices compared to non-waterfront homes, reflecting the scarcity and desirability of being in a waterfront property.

The interaction analysis between waterfront status and view quality did not produce a statistically significant interaction term. This suggests there is insufficient evidence to conclude that the effect of view quality differs between waterfront and non-waterfront property after controlling for the remaining predictors. However, both waterfront status and view quality independently contributed positively to housing prices.

The overall regression model explained a large proportion of the variability in housing prices, with an adjusted R^2 value near 0.685. This shows that approximately 68.5% of the variation in King County home prices can be explained by the selected predictors included in the model. The global F test was highly significant, providing strong evidence that at least one predictor variable contributes meaningfully to explaining housing prices.

Diagnostic procedures suggested that the assumptions of multiple linear regression were reasonably satisfied. Although some heteroscedasticity and influential outliers were present, these issues are common in housing datasets due to the existence of extremely expensive luxury properties. Given the large sample size, the regression model still provides reliable interpretation and useful insight into the factors affecting housing prices.

Overall, the analysis shows that structural housing quality, usable interior space, desirable location, and waterfront location are among the most important determinants of housing prices in King County.

## VII. Conclusion

This analysis examined which housing characteristics are most strongly associated with home sale prices in King County, Washington using multiple linear regression modeling. The result showed that several structural and location-based variables significantly contribute to explaining variation in housing prices.

Among the predictors considered, construction grade, interior living space, waterfront status, and distance from downtown Seattle were some of the most influential factors. Homes with higher construction quality, larger living areas, and desirable locations tended to have substantially higher sale prices. Bathrooms and view quality also showed significant positive relationships with housing prices, while lot size contributed relatively little after taking remaining predictors into account.

The regression model explained approximately 68.5% of the variability in housing prices, indicating that the selected predictors provided strong explanatory power. Although diagnostic analysis revealed some heteroscedasticity and influential observation, these issues were expected due to the presence of luxury and waterfront properties. Overall, the regression assumptions were reasonably satisfied and the model remained appropriate for interpretation and inference.

## Appendix

```
data <- read.csv("/Users/anishkadhar/Downloads/kc_house_data.csv")
str(data)
summary(data)
data <- subset(data, select = -c(id, date))
data <- data %>%
  mutate(
    waterfront = ifelse(is.na(waterfront), 0, waterfront),
    view = ifelse(is.na(view), 0, view),
    yr_renovated = ifelse(is.na(yr_renovated), 0, yr_renovated),
    renovated = ifelse(yr_renovated > 0, 1, 0),
    age = 2015 - yr_built,
    dist_downtown = sqrt((lat - 47.6062)^2 + (long + 122.3321)^2)
  )
dim(data)
hist(data$price,
     main = "House Price",
     xlab = "Price", breaks = 90)
hist(data$sqft_living,
     main = "Sqft Living",breaks = 90,
     xlab = "Price")
plot(data$sqft_living,
     data$price,
     main = "Price vs Living Area",
     xlab = "Living Area (sqft)",
     ylab = "Price",
     pch = 16,
     cex = 0.5)
abline(lm(price ~ sqft_living, data = data),
       lwd = 2)
plot(data$grade,
     data$price,
     main = "Price vs Grade",
     xlab = "Grade",
     ylab = "Price",
     pch = 16,
     cex = 0.6)
boxplot(price ~ waterfront,
        data = data,
        main = "Price by Waterfront Status",
        xlab = "Waterfront",
        ylab = "Price")
boxplot(price ~ view,
        data = data,
        main = "Price by View Rating",
        xlab = "View Rating",
        ylab = "Price")
boxplot(price ~ view,
        data = data,
```

```
        main = "Price by View Rating",
        xlab = "View Rating",
        ylab = "Price")

num_vars <- data[, sapply(data, is.numeric)]

cor_matrix <- cor(num_vars, use = "complete.obs")
round(cor_matrix, 3)
library(tidyverse)
library(MASS)
library(car)
library(broom)
library(lmtest)
library(lm.beta)
kc <- read.csv("kc_house_data.csv")
kc <- kc %>%
  mutate(
    waterfront = ifelse(is.na(waterfront), 0, waterfront),
    view = ifelse(is.na(view), 0, view),
    yr_renovated = ifelse(is.na(yr_renovated), 0, yr_renovated),
    renovated = ifelse(yr_renovated > 0, 1, 0),
    age = 2015 - yr_built,
    dist_downtown = sqrt((lat - 47.6062)^2 + (long + 122.3321)^2)
  )
main_model <- lm(price ~ bedrooms + bathrooms + sqft_living + sqft_lot +
                   floors + waterfront + view + condition + grade +
                   age + renovated + dist_downtown,
                 data = kc)
summary(main_model)
confint(main_model)
# Model selection
selected_model <- stepAIC(main_model, direction = "both", trace = FALSE)
summary(selected_model)
# Diagnostics
par(mfrow = c(2, 2))
plot(selected_model)

par(mfrow = c(1, 1))

vif(selected_model)
bptest(selected_model)

cook <- cooks.distance(selected_model)
plot(cook, type = "h",
     main = "Cook's Distance",
     ylab = "Cook's Distance")
abline(h = 4 / nrow(kc), col = "red")

kc_final <- kc[cook < 4 / nrow(kc), ]

final_model <- lm(formula(selected_model), data = kc_final)
summary(final_model)
confint(final_model)
# Correlation matrix
cor(kc_final[, c("price", "bathrooms", "sqft_living",
                 "sqft_lot", "grade", "age", "dist_downtown")])

# Summary statistics
```

```r
summary(kc_final[, c("price", "bathrooms", "sqft_living",
                     "sqft_lot", "grade", "age", "dist_downtown")])

# Export tables
write.csv(tidy(final_model, conf.int = TRUE),
          "final_model_coefficients.csv",
          row.names = FALSE)

write.csv(glance(final_model),
          "final_model_fit.csv",
          row.names = FALSE)
# Question 1: bathrooms
bath_model <- lm(price ~ bathrooms + bedrooms + sqft_living + grade +
                   condition + age + waterfront + view,
                 data = kc_final)
summary(bath_model)
confint(bath_model)
# Question 2: grade vs sqft living
grade_sqft_model <- lm(price ~ grade + sqft_living + bathrooms +
                         bedrooms + condition + age,
                       data = kc_final)
summary(grade_sqft_model)
lm.beta(grade_sqft_model)
# Question 3: distance from downtown
distance_model <- lm(price ~ dist_downtown + sqft_living + grade +
                       bathrooms + bedrooms + condition + age,
                     data = kc_final)
summary(distance_model)
confint(distance_model)
# Question 4: waterfront and view interaction
interaction_model <- lm(price ~ waterfront * view + sqft_living + grade +
                          bathrooms + bedrooms + condition + age +
                          dist_downtown,
                        data = kc_final)
summary(interaction_model)
confint(interaction_model)
# Question 5: structural traits vs age traits
structural_model <- lm(price ~ bedrooms + bathrooms + sqft_living +
                         sqft_lot + floors + grade + condition +
                         waterfront + view,
                       data = kc_final)

age_model <- lm(price ~ age + renovated, data = kc_final)

combined_model <- lm(price ~ bedrooms + bathrooms + sqft_living +
                       sqft_lot + floors + grade + condition +
                       waterfront + view + age + renovated,
                     data = kc_final)

summary(structural_model)$r.squared
summary(age_model)$r.squared
summary(combined_model)$r.squared
# Question 6: sqft living vs sqft lot
living_lot_model <- lm(price ~ sqft_living + sqft_lot + bathrooms +
                         bedrooms + grade + condition + age,
                       data = kc_final)
summary(living_lot_model)
```

```r
lm.beta(living_lot_model)
model_compare <- lm(price ~ sqft_living + sqft_lot,
                    data = kc_final)

summary(model_compare)
library(lm.beta)
lm.beta(model_compare)
summary(model_compare)
lot_model <- lm(price ~ sqft_living + sqft_lot +
                  grade + bathrooms + bedrooms +
                  condition + age,
                data = kc_final)

summary(lot_model)

library(lm.beta)
lm.beta(lot_model)
# Structural traits model
structural_model <- lm(price ~ grade + sqft_living +
                         bathrooms + bedrooms +
                         condition,
                       data = kc_final)

summary(structural_model)

# Age only model
age_model <- lm(price ~ age,
                data = kc_final)

summary(age_model)

# Full combined model
full_model <- lm(price ~ grade + sqft_living +
                   bathrooms + bedrooms +
                   condition + age,
                 data = kc_final)

summary(full_model)
```